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From General Doctor to Specialist: Enhancing Radiology Report Generation with Retrieval-Augmented Generation

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Abstract

We propose a multimodal Retrieval-Augmented Generation (RAG) framework for generating clinically accurate radiology reports from chest X-rays. Our study systematically evaluates similarity metrics for retrieval and the impact of negative sampling within the RAG pipeline. The approach extracts predicted findings and scores from a TorchXRayVision model, builds vector indices over MIMIC-CXR training reports, retrieves relevant neighbors using multiple strategies, and generates reports via a Large Language Model. In 16 controlled experiments, RAG consistently outperformed a non-retrieval baseline across both clinical (CheXbert and RadGraph) and linguistic (BERTScore, BLEU, METEOR, ROUGE-L) metrics. Moreover, adding explicit negative sampling at the prompt level consistently degrades performance, indicating that dissimilar reports confuse the LLM rather than provide useful guidance. Conceptually, RAG grounds a general-purpose LLM with precise, case-specific exemplars, steering it toward the specialized phrasing and clinical judgment of an expert radiologist.

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Keywords: Radiology Report Generation; Retrieval-Augmented Generation; Similarity Metrics; Negative Sampling; Multimodal Learning

1. Introduction

Radiology report generation is a critical component in the diagnostic and treatment processes of modern medicine. Radiology employs imaging modalities such as X-rays, CT scans, and MRI to visualize internal body structures for diagnostic and treatment purposes. The resulting reports serve as a vital bridge between raw imaging data and clinical decision-making [1].

A standard radiology report typically comprises two main sections: the **Findings** section, which provides an objective description of observations, and the **Impression** section, which offers clinical interpretation, summarizes key conclusions, and may recommend further diagnostic or therapeutic actions. Radiologists must process large volumes of images and distill them into precise, structured textual reports—a task complicated by time constraints, heavy

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workloads, and the risk of interpretive variability. Given the high stakes, inaccuracies or ambiguities in these reports can significantly compromise patient care [2, 3].

Advances in artificial intelligence, particularly in natural language processing (NLP) and multimodal learning, have enabled automated systems to enhance the efficiency and precision of radiology report generation. Methods that jointly model images and text have proven especially effective in medical domains [4]. However, standalone generative models often suffer from hallucinations, inconsistent clinical style, and limited domain-specific expertise. Modern Large Language Models (LLMs), while powerful, behave as generalists—akin to a "general doctor" applying broad knowledge without case-specific exemplars—leading to generic or clinically inappropriate outputs.

Retrieval-Augmented Generation (RAG) addresses these shortcomings by incorporating relevant examples from external knowledge sources, yielding more accurate, factually grounded, and stylistically consistent outputs than purely generative approaches [5]. In radiology, RAG grounds the generator in historical reports from similar cases, supplying specialist-like examples and institutional phrasing that steer outputs toward accurate, clinically appropriate reports [6].

In this work, we integrate a visual predictor, named TorchXRayVision [7] with a retrieval component and an LLM generator (gpt-4o-mini [8]) to create a multimodal RAG framework tailored for chest X-ray report generation. Our primary contributions are:

1. **Systematic evaluation of feature aggregation**: We comprehensively assess aggregation techniques for multi-view X-ray images (PA, AP, lateral, and LL views) on the MIMIC-CXR [9] dataset, identifying mean pooling as the optimal strategy.
2. **Comprehensive analysis of retrieval strategies and similarity metrics**: We evaluate Full vs. Partial Retrieval setups and compare four similarity metrics in the full-retrieval setting to maximize context relevance.
3. **Empirical study of vector encodings and negative sampling**: We systematically investigate two encoding schemes—non-binary probability vectors vs. binary thresholded vectors—and analyze the impact of explicit negative sampling (including dissimilar reports in the prompt) on generative quality.

2. Related Works

Automated radiology report generation is a critical area in medical AI, aiming to reduce radiologist workload and improve patient care. A key challenge is the generation of incomplete or hallucinatory content, particularly references to non-existent prior studies, which are common in datasets like MIMIC-CXR [9]. To address this, preprocessing strategies like CXR-ReDonE [10] use GILBERT [11] a BioBERT-based [12] token classifier, to remove prior references, creating a refined dataset that enhances the performance of retrieval-based models like CXR-RePaiR [13]. Early methods approached report generation as image captioning, using CNN-RNN architectures, later evolving to Transformer-based encoder-decoder models [14, 15] and hybrid or graph-based approaches [16] to incorporate clinical knowledge, though challenges in ensuring report completeness and consistency persist. The emergence of large language models (LLMs) and vision-language models (VLMs) has advanced the field, with domain-specific models like CheXagent [17], designed for chest X-ray interpretation, and few-shot learners such as LLaVA-Med [18] and Med-Flamingo [19], optimized for biomedical visual question answering (VQA). Retrieval-Augmented Generation methods reduce hallucinations by grounding outputs in external knowledge bases. CXR-RePaiR [13], for example, reframes generation as a retrieval task, using CLIP [20] to rank image-report similarities, with sentence-level retrieval improving output relevance. Advanced RAG frameworks further refine this approach: LaB-RAG [6] employs categorical labels as natural language descriptors, derived from logistic classifiers on frozen image embeddings (e.g., BioViL-T [21]), to enhance retrieval through label-based filtering and prompt integration for in-context learning, achieving performance comparable to supervised fine-tuning methods like CheXagent and CXR-Mate [22]. Similarly, MMed-RAG [23], a multimodal RAG system, improves factual accuracy across medical domains like radiology by using domain-aware retrieval, adaptive context selection to filter irrelevant documents, and preference fine-tuning to address cross-modality and overall alignment issues, enhancing report generation quality. To align with radiologist workflows, models like CXR-Mate incorporate multi-view images and longitudinal data from prior reports, using LoRA adaptations and section embeddings to improve diagnostic accuracy, even when prior studies are unavailable [22]. Recent advancements introduce AI agents for tool orchestration, such as MedRAX [24], a specialized framework for chest X-ray interpretation. MedRAX employs a ReAct loop with LLMs like GPT-4o to integrate tools for report generation (using SwinV2),

classification (DenseNet-121), grounding (Maira-2 [25]), and VQA (CheXagent, LLaVA-Med), improving complex reasoning compared to end-to-end models.

3. Methodology

This section describes the multimodal Retrieval-Augmented Generation (RAG) pipeline we use for chest X-ray report generation. We first outline the overall design and then provide a formal description of the indexing and retrieval operations, the two vector representation schemes (non-binary vs. binary), the similarity metrics evaluated, the retrieval modes, and the negative-sampling strategy.

3.1. Overview

Our objective is to generate clinically accurate FINDINGS and IMPRESSION sections for chest X-rays by combining (1) structured visual predictions from a radiology image model and (2) case-relevant textual exemplars retrieved from a large corpus of reports. The design is driven by two hypotheses: (i) retrieval of exemplar reports supplies stylistic and co-occurrence priors that reduce hallucinations and improve phrasing, and (ii) the way we represent visual outputs (continuous probabilities vs. thresholded binaries) and the similarity metric used for retrieval materially affects which exemplars are returned and therefore the final generation quality.

The complete pipeline is shown in Fig. 1 and consists of three stages:

1. Image-to-vector extraction via TorchXRayVision
2. Retrieval over an indexed corpus of training reports
3. Conditioned generation with gpt-4o-mini

We evaluate multiple variants of stage (2) while keeping (1) and (3) fixed to isolate the effects of vector encoding, distance metric, retrieval granularity, and negative-sampling.

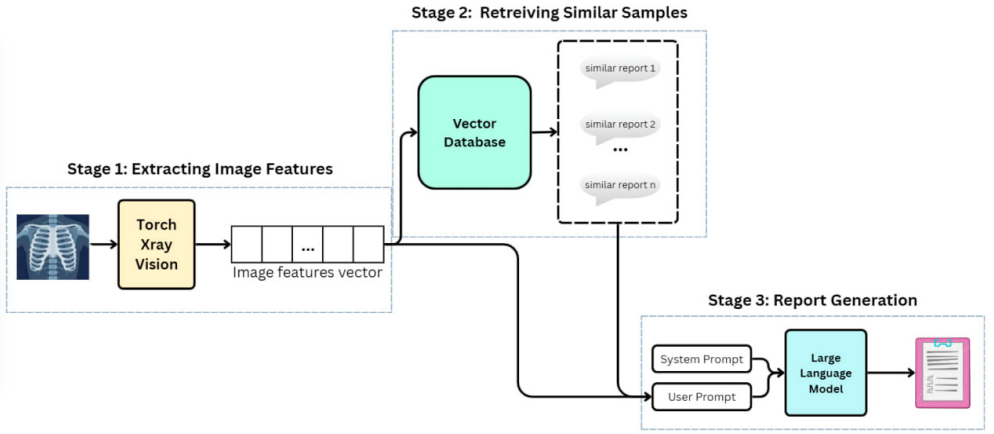


Fig. 1: Proposed pipeline for chest X-ray report generation, consisting of (1) Extracting features from multi-view images, (2) Retrieving similar samples from the MIMIC-CXR training corpus using the selected similarity metric, and (3) Report generation with gpt-4o-mini

3.2. Dataset and Preprocessing

We build a retrieval index from the MIMIC-CXR training portion with the studies that contain both FINDINGS and IMPRESSION. TorchXRayVision outputs a 14-dimensional probability vector per view, corresponding to the standard CheXbert pathologies. After aligning with the textual labels present in the MIMIC-CXR reports and excluding dataset-specific or unsupported classes, **11 pathologies remain fully overlapping**. All subsequent retrieval indexing,

similarity computations, and clinical evaluations are conducted exclusively within this shared 11-dimensional label space.

3.3. Formal notation

Let a study s contain m image views. For each view $v \in \{1, \ldots, m\}$, the visual model outputs a probability vector

$$\mathbf{p}^{(v)} \in [0,1]^d,$$

where d is the number of label classes. We aggregate view-level outputs into a single *study vector* using mean pooling:

$$\mathbf{p}_s = \frac{1}{m} \sum_{v=1}^{m} \mathbf{p}^{(v)}.$$

We also construct a binary variant by thresholding:

$$\mathbf{b}_s[i] = \mathbf{1}\{\mathbf{p}_s[i] \geq \tau\},$$

where τ is a chosen threshold. Define the retrieval corpus (index) as

$$\mathcal{D} = \{(\mathbf{x}_j, r_j)\}_{j=1}^{N},$$

where $\mathbf{x}_j$ is the stored vector for training study j (either $\mathbf{p}_j$ or $\mathbf{b}_j$) and r_j is the textual report. Given a query study s with query vector $\mathbf{x}_q$, we compute a similarity score $\mathrm{sim}(\mathbf{x}_q, \mathbf{x}_j)$ for every j and return the top-K reports ranked by similarity. The end-to-end process for a single test study s is summarized below in pseudo-code 1:

Algorithm 1: Multi-View Retrieval-Augmented Report Generation

Input : study images, index $\mathcal{D}$, similarity metric sim, threshold τ, retrieval mode, K, T, K_{neg}

Extract per-view probabilities $\{\mathbf{p}^{(v)}\}_{v=1}^{m}$
$\mathbf{p}_s \leftarrow \frac{1}{m} \sum_v \mathbf{p}^{(v)}$
if *binary mode* **then**
  $\mathbf{b}_s \leftarrow \mathbf{1}\{\mathbf{p}_s \geq \tau\}$
  $\mathbf{x}_q \leftarrow \mathbf{b}_s$
end
else
  $\mathbf{x}_q \leftarrow \mathbf{p}_s$
end
if *Full retrieval* **then**
  Compute $\mathrm{sim}(\mathbf{x}_q, \mathbf{x}_j)$ for all j and return top-K reports
end
else
  Select top-T indices of $\mathbf{p}_s$
  Perform per-label masked retrieval, return aggregated set of up to K reports
end
Optionally add K_{neg} farthest reports as negative exemplars
Construct user prompt with predicted findings (with/without scores), positive similar samples (and negatives)
Call LLM to generate FINDINGS and IMPRESSION

3.4. Similarity metrics

We evaluate four common metrics for measuring proximity between study vectors: **Cosine**, **Euclidean**, **Jaccard**, and **Hamming**. Cosine similarity emphasizes the *direction* of the feature vectors rather than their magnitude, making it suitable when the relative pattern of findings is more informative than their absolute probabilities. Euclidean distance captures overall geometric closeness in the feature space, rewarding cases with similar probability magnitudes across all labels. Jaccard similarity focuses on the overlap of activated findings and is therefore effective for binary representations that describe presence or absence of conditions. Finally, Hamming similarity measures how many individual label predictions match between two studies, which helps identify reports with nearly identical binary finding profiles.

This choice deliberately spans the most common geometric assumptions in medical retrieval: Cosine and Euclidean respect probability magnitudes and diagnostic confidence in continuous space, whereas Jaccard and Hamming enforce

strict binary agreement on presence/absence of findings—allowing us to determine whether preserving rich probabilistic information or collapsing it to binary labels better drives clinically accurate and fluent report generation.

3.5. Retrieval modes

We compare two complementary retrieval granularities to examine the classic global-vs-local trade-off in medical exemplar selection:

Full retrieval. Compute similarity between the full study vector $\mathbf{x}_q$ and every $\mathbf{x}_j \in \mathcal{D}$, and return the top-$K$ reports $\{r_{(1)}, \ldots, r_{(K)}\}$. This mode retrieves examples that are globally similar across the full label set, favoring studies that match overall disease co-occurrence patterns.

Partial (per-finding) retrieval. Identify the top-T labels by descending probability in $\mathbf{p}_s$. For each selected label index i, perform a masked retrieval that focuses on that label. Aggregate the per-label neighbors into a short exemplar set. This mode emphasizes fine-grained exemplars for the most likely findings.

Both modes are evaluated to determine which strategy better supports clinically accurate and linguistically fluent report generation.

3.6. Negative-sampling design

To study negative guidance, we construct prompt contexts that contain:

- K_{pos} positive exemplars (top-K similar reports), and
- K_{neg} negative exemplars (the most dissimilar reports by the chosen similarity metric).

Negative examples are explicitly labeled in the prompt as "negative" and instructions state that the model should avoid making the errors illustrated by these negative exemplars. This is a prompt-level test of negative sampling (distinct from contrastive training).

3.7. Prompt design

Our framework uses two levels of prompting: a fixed **system prompt** that defines the model's role and behavior, and a dynamically generated **user prompt** that provides case-specific input. The system prompt instructs the LLM to act as a board-certified radiologist and to produce only the *FINDINGS* and *IMPRESSION* sections. We design four variants of the system prompt to match the experimental configurations:

1. A baseline version that relies solely on TXR-predicted labels without any retrieved reports
2. A standard RAG version that instructs the model to also use retrieved reports as additional context
3. A partial-retrieval version that highlights using reports retrieved for each individual finding along with TXR labels
4. A version used in experiments with both similar and dissimilar reports, where the model is told to use similar reports for guidance and negative (dissimilar) ones only to avoid incorrect inferences

The **user prompt** is generated dynamically for each study and contains the predicted findings (with or without their probability scores) and the retrieved positive or negative reports depending on the experimental setup. The system prompt used in most retrieval-augmented experiments is shown in Table 1, while the user prompt template is presented in Table 2.

4. Experiments

4.1. View aggregation

Each radiology study in MIMIC-CXR may contain multiple image views. To determine the optimal way to aggregate per-view predictions into a single study-level vector, we compared two common strategies: **maximum pooling** and

Table 1: System prompt used in most RAG experiments

```
You are an expert radiologist specializing in chest X-rays.
Your task is to generate only the FINDINGS and IMPRESSION sections of a chest X-ray
report.
Follow these rules:
- Base your report on the provided FINDINGS and retrieved similar reports for guidance.
- Write in a concise, professional tone used in real radiology reports.
- Do NOT copy text verbatim from retrieved reports unless medically appropriate.
- Do NOT mention any numeric scores, probabilities, confidence values, or thresholds in
the report text.
- Do NOT include patient identifiers, clinical history, or section headers other than
FINDINGS and IMPRESSION.
- Use complete sentences and standard medical terminology.
```

Table 2: User prompt used in all RAG experiments (except the baseline)

```
Findings:
{predicted_findings}

Retrieved similar reports:
{similar_reports}

Generate a new FINDINGS and IMPRESSION section for this case.
```

mean pooling. Each approach combines the per-view probability vectors output by TorchXRayVision into a single d-dimensional representation before retrieval. We evaluated these aggregation methods on the validation subset using micro-F1 as the clinical metric. As shown in Table 3, mean pooling with $\tau = 0.65$ achieved the highest performance and was therefore used in all subsequent experiments. We further evaluated the optimal threshold (0.65) on the held-out test set and observed that it also yields the highest micro-F1 score there. This consistent performance across validation and test sets indicates that the threshold is robust and generalizes effectively to unseen data.

Table 3: Comparison of micro-F1 scores on the validation set for TXR using average vs. max pooling across different confidence thresholds

	0.40	**0.45**	**0.50**	**0.55**	**0.60**	**0.65**	**0.70**	**0.75**	**0.80**	**0.85**	**0.90**	**0.95**
Average pooling	61.17	64.35	66.55	80.47	83.66	**84.10**	83.44	82.34	80.81	79.35	78.04	76.86
Max pooling	42.48	44.32	45.98	74.73	80.64	82.59	**82.98**	82.45	81.32	79.71	78.25	76.92

4.2. Experimental Configurations

We ran a set of 16 controlled experiments varying:

- Vector type: with scores (non-binary) vs. without scores (binary).
- Similarity metric: Cosine, Euclidean, Jaccard, Hamming.
- Retrieval mode: Full vs. Partial.
- Negative sampling: absent vs. present.

5. Results

We evaluated the FINDINGS and IMPRESSION sections separately. Results are shown in Tables 4 and 5, respectively. Key findings from controlled ablations are as follows:

Table 4: Evaluation results for the FINDINGS section on the test set. Higher is better in all metrics.
Experiment abbreviations: w/ = with probability scores of predicted findings, **w/o** = without probability scores of predicted findings, **+Neg** = with negative examples

#	Experiment	BERTScore	BLEU	METEOR	ROUGE-L	CheXbert	RadGraph
1	Baseline	0.444	0.016	0.185	0.137	0.505	0.103
2	Random	0.519	0.030	0.247	0.161	0.515	0.191
3	Full + Cosine + w/	**0.521**	0.029	0.249	0.161	**0.525**	0.199
4	Full + Cosine + w/o	0.509	0.029	0.237	0.158	0.519	0.181
5	Full + Euclidean + w/	**0.521**	0.030	0.248	**0.162**	0.523	**0.202**
6	Full + Euclidean + w/o	0.509	0.029	0.236	0.158	0.520	0.182
7	Full + Hamming + w/	**0.521**	0.029	0.249	**0.162**	0.524	0.199
8	Full + Hamming + w/o	0.508	0.029	0.237	0.158	0.519	0.179
9	Full + Jaccard + w/	**0.521**	0.030	0.249	**0.162**	0.524	0.199
10	Full + Jaccard + w/o	0.508	0.029	0.237	0.158	0.518	0.179
11	Full + Hamming + w/ + Neg	0.511	0.025	0.234	0.156	0.522	0.191
12	Full + Jaccard + w/o + Neg	0.511	0.025	0.235	0.156	0.523	0.192
13	Partial + Simple + w/	0.518	0.029	0.246	0.160	0.515	0.194
14	Partial + Simple + w/o	0.504	0.027	0.234	0.157	0.516	0.174
15	Partial + Relevant + w/	0.513	**0.033**	**0.263**	0.161	0.509	0.200
16	Partial + Relevant + w/o	0.503	**0.033**	0.256	0.160	0.515	0.185

Table 5: Evaluation results for the IMPRESSION section on the test set. Higher is better in all metrics.
Experiment abbreviations: w/ = with retrieved reports, **w/o** = without retrieved reports, **+Neg** = with negative examples

#	Experiment	BERTScore	BLEU	METEOR	ROUGE-L	CheXbert	RadGraph
1	Baseline	0.313	0.004	0.147	0.074	0.359	0.077
2	Random	0.380	0.010	0.187	0.105	0.352	0.111
3	Full + Cosine + w/	0.391	**0.011**	0.195	0.108	0.365	0.118
4	Full + Cosine + w/o	0.363	0.008	0.170	0.090	0.363	0.100
5	Full + Euclidean + w/	0.389	**0.011**	0.195	0.107	0.360	**0.119**
6	Full + Euclidean + w/o	0.363	0.007	0.172	0.094	0.365	0.102
7	Full + Hamming + w/	**0.392**	0.010	0.196	**0.109**	**0.366**	0.118
8	Full + Hamming + w/o	0.362	0.009	0.168	0.093	0.362	0.098
9	Full + Jaccard + w/	0.390	0.010	0.195	0.108	0.364	0.116
10	Full + Jaccard + w/o	0.363	0.007	0.169	0.093	0.361	0.097
11	Full + Hamming + w/ + Neg	0.376	0.010	0.189	0.106	0.362	0.115
12	Full + Jaccard + w/o + Neg	0.375	0.010	0.186	0.104	0.361	0.114
13	Partial + Simple + w/	0.384	0.010	0.190	0.105	0.357	0.112
14	Partial + Simple + w/o	0.359	0.007	0.165	0.090	0.360	0.097
15	Partial + Relevant + w/	0.372	0.008	**0.207**	0.106	0.350	**0.119**
16	Partial + Relevant + w/o	0.349	0.007	0.188	0.095	0.360	0.107

1. **Effect of RAG:** Retrieval-Augmented Generation substantially improves both clinical metrics and language quality metrics compared to the no-retrieval baseline.
2. **Similarity metrics:** Euclidean and Cosine measures yield the most consistent performance across both clinical and linguistic metrics. Binary-only metrics (Jaccard, Hamming) perform competitively in some language metrics but underperform clinically.
3. **Vector representations:** Non-binary (probability) vectors outperform binary encodings. The richer continuous probabilities allow the retriever to capture nuanced finding severity and co-occurrence patterns, leading to more accurate and realistic reports.
4. **Negative sampling:** Including explicit dissimilar (negative) reports in the context reduces performance, confirming that contradictory examples at the prompt level confuse the LLM rather than guide it.

5. **Retrieval mode:** Full retrieval achieves the best overall clinical accuracy, while partial retrieval can enhance detailed description of top findings but may miss less probable conditions.

6. Conclusion

We presented a systematic study of retrieval strategies for radiology report generation, analyzing the impact of similarity metrics, vector encodings, and negative sampling. Our experiments demonstrate that Retrieval-Augmented Generation significantly improves clinical accuracy and linguistic quality by grounding a generalist LLM with relevant exemplar reports—helping it behave more like a specialist radiologist.

Among all tested configurations, non-binary probability vectors combined with Euclidean or Cosine similarity in Full Retrieval yielded the best clinical results. Binary encodings can be sufficient for lightweight retrieval but lose subtle probability information crucial for clinical phrasing. Moreover, naïve prompt-level negative sampling consistently reduces performance, confirming that simply including dissimilar reports in the prompt confuses the LLM rather than guiding it.

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